Exercise 2

Table Employees

ID	First Name	Last Name	Department	Salary	Hire-date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

Questions

1 Count ( ) Function

Write SQL query to ~~count~~ find total number of employees in the company

```
SELECT COUNT(*) AS total employees
  FROM employees;
```

2. SUM ( ) function

Write SQL query to find total salary paid to all employees in the IT department

```
SELECT SUM(salary) AS total_it_salary
  FROM employees
   WHERE department = 'IT';
```

3. AVG() function

Write a SQL to calculate the average salary of employees in the HR department

```
SELECT AVG(salary) AS average_HR_salary
  FROM employees
   WHERE department = 'HR';
```

4. MIN() and MAX() functions

SQL

```
SELECT MIN(salary) AS lowest_salary,
       MAX(salary) AS highest_salary
FROM employees;
```

5 GROUP BY Statement

SQL

```
SELECT department, SUM(salary) AS total_salary
 FROM employees
  GROUP BY department;
```

6 GROUP BY and Count()

Write SQL query to count how many employees work in each in each city

```
SELECT city, Count(*) AS employee_count
 FROM employees
  GROUP BY city;
```

7. GROUP BY and ORDER BY

Group by department, calculate average salary, order by average salary descending

```
SELECT department, AVG(salary) AS average-salary
FROM employees
GROUP BY department
ORDER BY average-salary DESC;
```

8. Having Clause

Write SQL query to find departments where the total salary paid exceeds 100,000 (Use GROUP BY & HAVING)

```
SELECT department, SUM(salary) AS total salary
FROM employees
GROUP BY department
Having SUM(salary) > 100 000;
```

9. Combining GROUP BY, Having and ORDER BY

List cities where more than one employee works, ordered by number of employees descending

```
SELECT city, Count (*) AS employees
FROM employees
GROUP BY city
Having Count (*) > 1
ORDER BY employee_count DESC;
```

10 Combining Aggregate functions
Find department with highest average salary

SQL

```
SELECT department, AVG(salary) AS average_salary
 FROM employees
  GROUP BY department
   ORDER BY average_salary DESC
    Limit 1;
```

Exercise 2 Table outcomes

Question 1

Count ()

total_employees
10

Question 2

SUM() (IT department)

total_salary
22 000

Question 3

AVG() HR department
average_hr_salary
49 500

Question 4

MIN() and MAX()

lowest salary	Highest salary
48000	62 000

Question 5

GROUP BY (department_Totals)

department	total_salary
IT	22 0000
HR	99000
Finance	119000
Marketing	105000

Question 6

Employees per city

city	employee_count
New York	2
Chicago	3
Los Angeles	2
San francisco	2
Houston	1

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Question 7

AVG Salary per Department

department	AVG_salary
Finance	59500
IT	55000
Marketing	52500
HR	49500

Question 8

Departments with Total_salary > 100000

department	total_salary
IT	220000
Finance	119000
Marketing	105000

Question 9

Cities with more than 1 employee

City	employee_count
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

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Question 10

Highest AVG salary department

department	average salary
Finance	59500